



# Transformer-based multimodal learning for predicting mechanical properties in heat-treated stainless steel

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## ABSTRACT

Accurately predicting mechanical properties of heat-treated materials is critical for intelligent process control and advanced manufacturing. This study proposes a Transformer-based multimodal learning framework for predicting the hardness and wear behavior of carburized steel after vacuum carburizing. By integrating microstructural images, material compositions, and process parameters, the proposed model effectively captures complex cross-modal relationships. Experimental results show that the multimodal model achieves high prediction accuracy, with an  $R^2$  of 0.98 and MAE of 5.23 HV for hardness prediction. Furthermore, Variational Mode Decomposition (VMD) is introduced to preprocess the wear curve, reducing noise and improving the robustness of friction performance prediction. The results demonstrate the effectiveness and generalizability of the proposed approach, offering a practical AI-based solution for intelligent material property evaluation and process optimization.

## 1. Introduction

Carburized steels are widely used in industrial fields such as automotive, aerospace, and manufacturing due to their high strength, high hardness, and excellent wear resistance. To further enhance their surface properties, vacuum carburizing—a thermochemical treatment technique—has been extensively adopted, offering precise control over carbon diffusion depth while effectively suppressing surface oxidation [1,2]. However, accurately predicting the mechanical properties (e.g., hardness gradient and wear resistance) of the material after such heat treatment remains a significant challenge, owing to the complex and nonlinear coupling among alloying elements, microstructural evolution, and heat treatment parameters such as temperature and holding time [3].

Traditional approaches to property prediction rely heavily on empirical models or extensive experimental trials [4–6]. While these methods can provide insights under controlled settings, they are often time-consuming, expensive, and lack scalability across different alloy systems or processing routes. Moreover, they struggle to account for the

intrinsic multi-physics interactions—such as the influence of carbon content on phase transformation or the impact of process parameters on residual stress and carbide morphology. In recent years, data-driven approaches based on machine learning have gained momentum in materials science, including phase classification [7,8], grain size estimation [9], and dual-phase steel recognition [10]. However, these methods often rely on only one modal data (e.g., images or composition alone), limiting their ability to capture the full complexity of carburized steels where mechanical properties result from coupled effects among structure, chemistry, and processing [11,12]. In addition to conventional and data-driven models, recent studies have explored physics-informed neural networks (PINNs) [13–15], wavelet-based learning methods [16,17], and transfer learning [18–20] for mechanical property prediction in solid materials. PINNs embed physical laws into neural networks to improve interpretability and consistency; wavelet transforms enable multiscale feature extraction from non-stationary signals such as wear curves; and transfer learning allows model adaptation with limited data. However, most of these approaches rely on single-modal inputs and lack holistic integration of structural, compositional, and process

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features.

To overcome these limitations, multimodal learning has emerged as a promising strategy. Prior work has demonstrated its potential in various materials contexts—for example, Ren et al. [21] proposed a multimodal model for predicting the tensile properties of dual-phase steels, while Leong et al. [22] introduced multimodal language models for mining electrosynthesis reactions. Despite its growing success in various materials contexts, multimodal learning remains underexplored in surface treatment applications—especially for vacuum-carburized systems where gradient structures pose unique modeling challenges. Surface engineering requires comprehensive modeling of complex microstructure–processing–property relationships, including interface effects and the synergistic behavior between hardened layers and substrates. Addressing this gap, we propose a customized multimodal learning framework specifically tailored to predict the post-treatment mechanical properties of vacuum-carburized steels.

The proposed framework integrates three primary data modalities: metallographic micrographs, chemical composition, and carburizing process parameters. We adopt a hierarchical attention-based Transformer architecture to enable fine-grained fusion of heterogeneous features [23–25]. Specifically, a Vision Transformer (ViT) [26] is employed to extract spatial features from microstructural images, while parallel encoders process compositional and process-related numerical data. These representations are then fused in a multimodal encoder that facilitates joint reasoning across modalities. To further enhance prediction robustness, especially for wear-related behaviors, we incorporate Variational Mode Decomposition (VMD) to preprocess the friction coefficient signals. VMD extracts physically meaningful intrinsic modes from high-dimensional time series, effectively denoising raw input and improving the model's capacity to recognize wear patterns and underlying physical trends [27,28]. The main contributions are as follows:

- (1) We proposed a novel Transformer-based architecture tailored for the multimodal fusion of microstructural, compositional, and process parameters to predict the mechanical properties of carburized steels.
- (2) VMD is introduced to enhance signal interpretability and prediction stability in wear analysis.
- (3) Through systematic comparison with conventional CNN-based architectures and state-of-the-art multimodal learning strategies, our proposed approach consistently outperforms baseline methods, achieving an  $R^2$  of up to 0.98 and a MAE as low as 5.23 HV in hardness prediction.
- (4) A comprehensive experimental validation is conducted across various combinations of compositions and processing parameters.

By effectively leveraging the complementary strengths of different data modalities, this framework offers a powerful and scalable AI-based solution for mechanical property prediction and intelligent process design in surface engineering.

## 2. Proposed Methods

### 2.1. Background and framework

The mechanical properties of carburized steel after vacuum carburizing are influenced by various factors, including material composition, structure, and processing. In this work, we focus on the relationships between material composition, microstructure, heat treatment process, and material properties. Material composition serves as the foundation for sample preparation and is readily obtainable; microstructure typically requires specialized equipment for acquisition, with metallographic images reflecting features such as crystal size, shape, surface texture, porosity, and material defects, all of which are closely related to material properties [29]. Traditional methods of obtaining properties

through physical experiments, as shown in Fig. 1(a), are both time-consuming and uneconomical. With the advancement of artificial intelligence technologies, data-driven approaches are gaining increasing attention [21,30–36]. Many studies have developed mapping models for property prediction based on material composition or microstructure, but they have not comprehensively considered the impact of multiple inputs on the final properties, resulting in less complete mapping relationships [37–39].

To address this, we adopt a multimodal learning strategy that integrates composition, microstructure, and vacuum carburizing process data to construct a more holistic and accurate predictive model for mechanical properties. A multimodal database was established, comprising diverse compositions, processing routes, and metallographic images. As shown in Fig. 1(b), we designed a Transformer-based architecture to effectively fuse visual and tabular data, enabling both unimodal understanding and multimodal reasoning. The model framework, illustrated in Fig. 2, consists of an image encoder, a text encoder, a multimodal encoder, and a multilayer perceptron for regression. Detailed formulations of feature embedding, modality fusion, and prediction modules are provided in the [Supplementary Material](#). In our model, microstructural information is used solely in the form of metallographic images, not as manually extracted features like phase fractions or grain size. These images are treated as a visual modality and processed by a Vision Transformer to automatically learn structural patterns. This design reflects practical workflows where image data are directly obtained after processing and avoids violating the causal relationship between composition, process, microstructure, and properties.

#### 2.1.1. Microstructure encoder

The microstructure encoder adopts the ViT (Vision Transformer) architecture. The input microstructural images of carburized steel after vacuum carburizing are preprocessed to a fixed size of  $224 \times 224$  and divided into non-overlapping  $16 \times 16$  patches, resulting in 196 patches per image. Each patch captures local metallographic features such as phase distribution, grain boundaries, and carbide morphology. Each patch is flattened and mapped to a 768-dimensional embedding vector, followed by the addition of learned positional encodings. A special classification token  $[cls]$  is prepended to the sequence to aggregate global image information. The embedded sequence is then fed into a Transformer encoder consisting of 12 layers, each with 12 attention heads, an intermediate hidden size of 3072, and dropout probability set to 0.1. After multiple layers of self-attention and feedforward processing, the encoder outputs the hidden states of all patches  $\{h_i\}$ , along with the hidden state corresponding to  $[cls]$ , denoted as  $h_{cls,i}$ , which serves as the global representation of the input microstructural image. Parameter Encoder.

We fix the material composition, physical parameters, and corresponding process route parameters at specific positions within a predefined list and concatenate this structured information into a single sequence. A standard tokenization method is then applied to divide this sequence into basic textual tokens, each of which is mapped to a fixed-dimensional word embedding, forming an input sequence of token vectors. This sequence is fed into a Transformer encoder comprising 12 layers, each with 12 attention heads, a hidden size of 768, and an intermediate feedforward size of 3072. The dropout probability is set to 0.1 in this implementation. Through multiple layers of self-attention and feedforward computation, the encoder generates a sequence of hidden states  $\{h_T\}$ . Within this sequence, a special classification token  $[cls_T]$  is inserted at the beginning and processed along with the other tokens. Its corresponding hidden state  $h_{cls,T}$  is used to aggregate the global semantic representation of the entire input. This enables the model to capture and integrate the intrinsic relationships between material composition, physical properties, and process parameters. The parameter encoder adopts the same ViT-style Transformer architecture as the microstructure encoder, promoting architectural consistency and facilitating effective fusion of multimodal information.

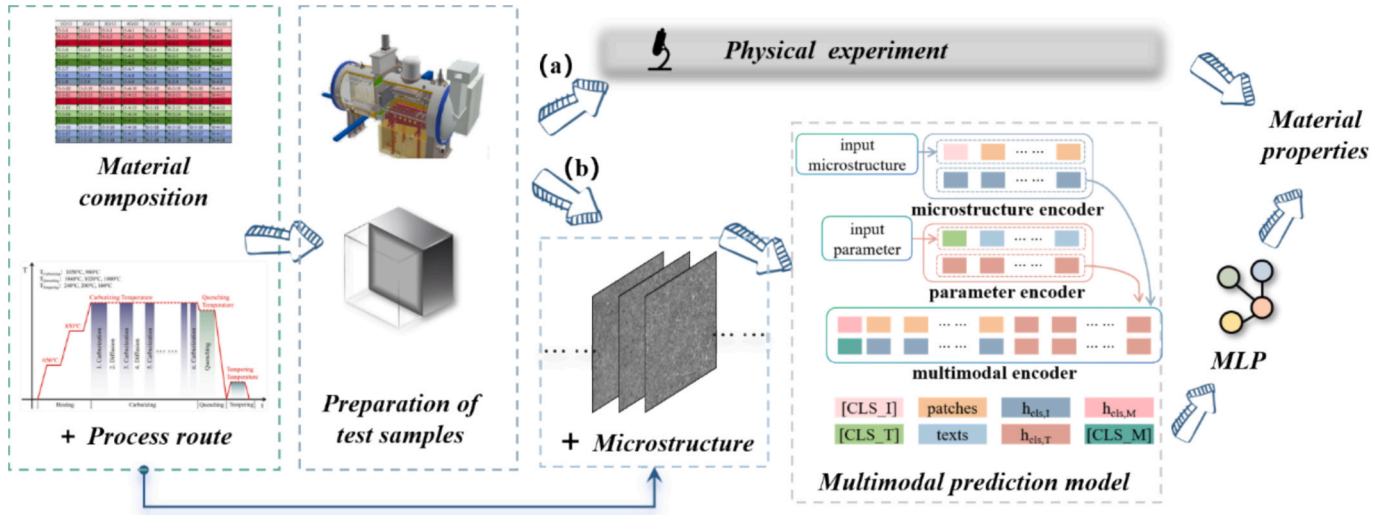


Fig. 1. Workflow. (a) mechanical properties measurement. (b) multimodal prediction.

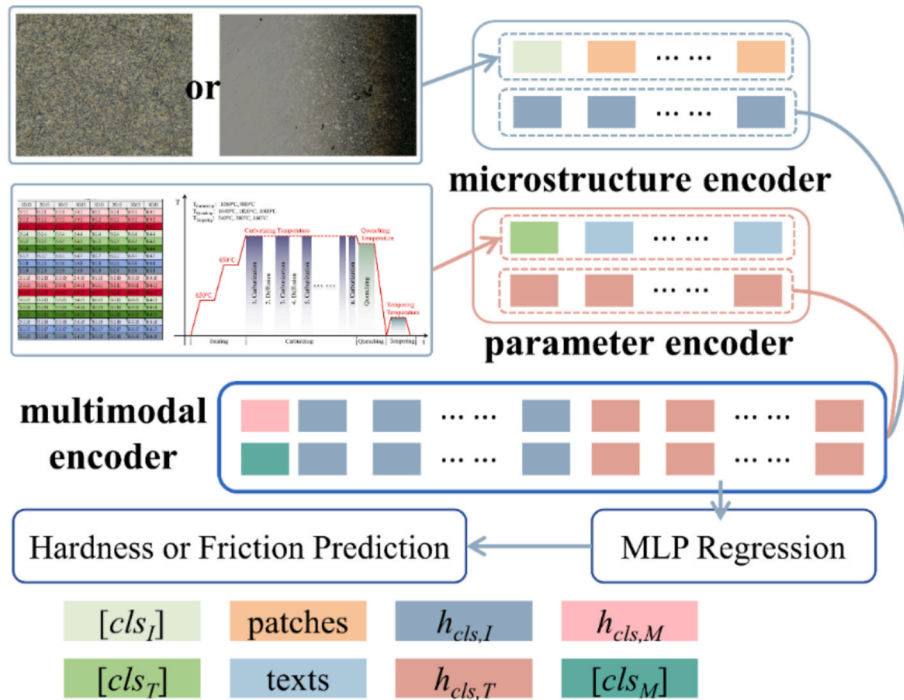


Fig. 2. Network architecture.

### 2.1.2. Multi-modal Encoder

We employed an independent Transformer model to achieve the fusion of the hidden states of metallographic images and experimental material process parameters. First, we apply two learnable linear projections to the hidden states  $\{h_I\}$  output by the microstructure encoder and  $\{h_T\}$  from the parameter encoder, respectively, mapping the representations from different modalities into a shared feature space. The projected image and parameter tokens are then concatenated into a single vector sequence, with a special classification token  $[cls_M]$  prepended. This multimodal vector sequence is fed into a Transformer encoder based on the ViT architecture, where multi-head self-attention enables cross-modal interaction and joint representation learning between metallographic image features and process parameter information. The multimodal encoder consists of 6 Transformer layers, each with 12 attention heads, a hidden size of 768, an intermediate

feedforward size of 3072, and a dropout probability of 0.1. After processing, the model outputs a sequence of multimodal hidden states  $\{h_M\}$ , where each vector encodes the fused features resulting from cross-modal integration of the original unimodal representations.

### 2.1.3. Regression Prediction

We utilize a multilayer perceptron (MLP) to perform regression prediction on the output of the multimodal encoder. Specifically, we extract the hidden state vector  $h_{cls,M}$  corresponding to the special token  $[cls_M]$  from the output of the multimodal Transformer encoder, which serves as a global representation of the fused cross-modal features from metallographic images and process parameters. This vector is then fed into an MLP composed of several fully connected layers. Through a series of transformations and nonlinear activations, the model produces a set of continuous outputs representing the predicted target variables,

such as the hardness and wear curve of carburized steel after vacuum carburizing. The parameters of the MLP are optimized using back-propagation and gradient descent, with the mean squared error (MSE) between the predicted and true values used as the loss function. This enables the model to effectively learn the regression mapping from fused multimodal representations to quantitative material properties.

## 2.2. Wear Curve Processing Method Based on VMD

Unlike the hardness curve that describes the hardness gradient of a carburized cross-section, the wear curve exhibits greater complexity. Directly using the entire friction coefficient curve as learning data can result in a very large output space for the model, increasing the difficulty of model training and leading to overfitting. Additionally, many data points in the friction coefficient curve are highly correlated and contain redundant information, which does not require complete prediction. In this study, the VMD technique is employed to decompose the complex friction coefficient curve into two modes with different frequency characteristics, thereby transforming the high-dimensional output into high-frequency and low-frequency modal outputs.

Mathematically, VMD decomposes a given signal  $f(t)$  into a finite number of  $K$  band-limited intrinsic mode functions (IMFs)  $\{u_k(t)\}$ , each associated with a center frequency  $\omega_k$ , by solving the following constrained variational problem:

$$\min_{\{u_k\}, \{\omega_k\}} \left\{ \sum_{k=1}^K \left\| \partial_t \left[ \left( \delta(t) + \frac{j}{\pi t} \right) * u_k(t) \cdot e^{-j\omega_k t} \right] \right\|_2^2 \right\} \quad \text{s.t.} \quad \sum_{k=1}^K u_k(t) = f(t) \quad (1.1)$$

Here, the objective is to minimize the sum of the bandwidths of all modes in the frequency domain (via Hilbert transform), while ensuring that the sum of all modes reconstructs the original signal  $f(t)$ . This optimization is typically solved using an iterative Alternate Direction Method of Multipliers (ADMM).

For non-stationary and nonlinear curves such as friction force signals and vibration signals, traditional Fourier or wavelet transforms often fail to capture localized frequency information. In contrast, VMD adaptively decomposes the signal into several physically meaningful modes, each containing specific frequency components. Unlike Empirical Mode Decomposition (EMD), VMD avoids mode mixing, resulting in more stable and reliable decompositions. In our implementation, the friction coefficient curve is decomposed into two IMFs—one capturing the low-frequency mode  $u_{\text{low}}(t)$ , and the other the high-frequency mode  $u_{\text{high}}(t)$ ,

which serve as two separate prediction targets for the neural network.

As shown in Fig. 3, the low-frequency component is distributed around 0 Hz. Due to the high carbon content and hardness (700–800 HV) of the sample steel surface after vacuum carburization, the low-frequency mode reflects the gradual wear process of the surface layer, where the substrate bears more friction, leading to an overall increase in wear. In contrast, the high-frequency mode is concentrated at 130 Hz and 450 Hz with smaller peaks, reflecting the microstructural damage of the carburized surface, crack propagation, and instantaneous changes in contact points between the friction pairs.

## 3. Experiment and analysis

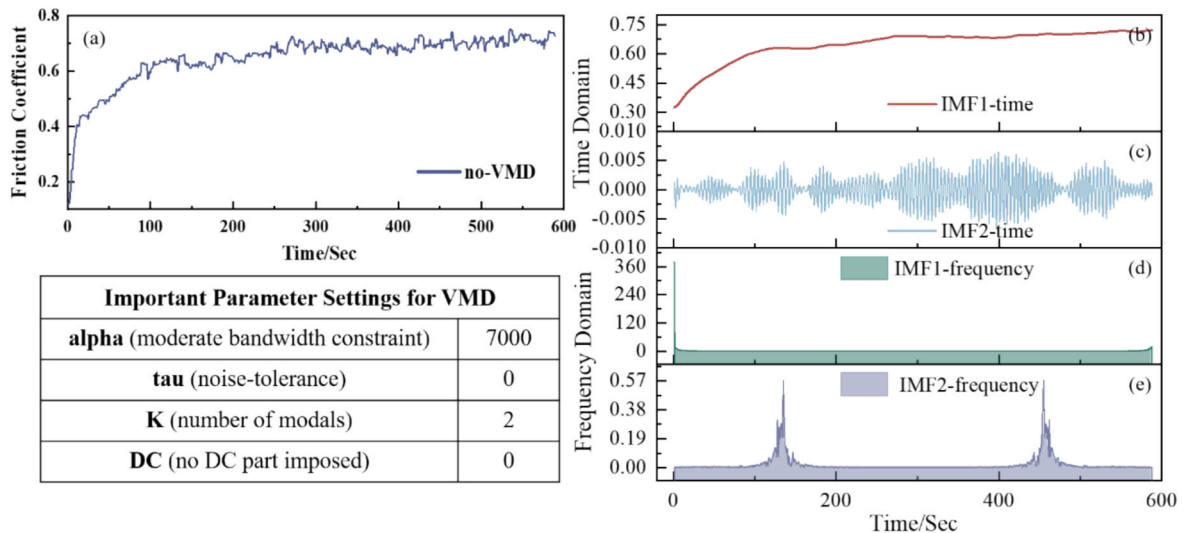
### 3.1. Experimental Setup and Process Parameters

In carburized steels, the surface quality and gradient of the carburized layer determine the fatigue and wear on the specimen surface. In this study, different types of carburized steel with different carbon contents were used to validate the proposed method. All materials were delivered in an annealed state, the material compositions are shown in Table 1.

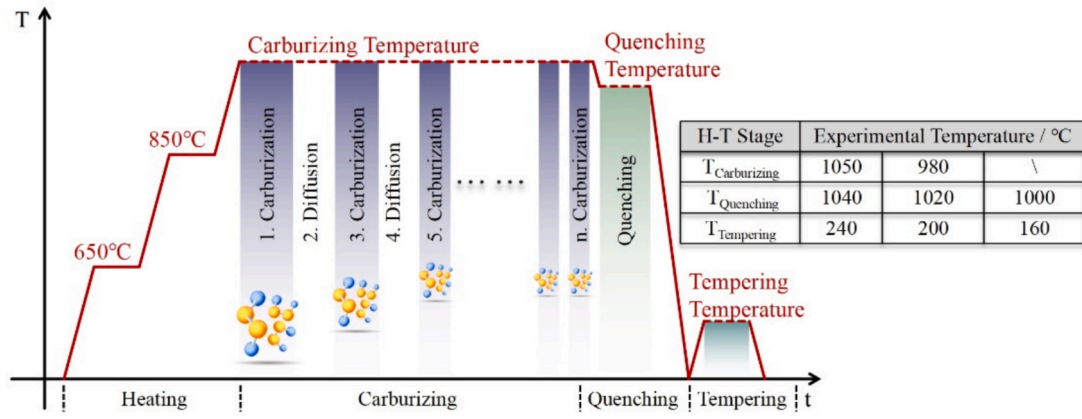
Martensitic transformation typically occurs within a very short time frame and is primarily governed by temperature rather than time. However, traditional atmospheric carburization often induces surface oxidation, compromising the uniformity and integrity of the carbide layer. To address this, we adopted a low-pressure vacuum carburization method using acetylene ( $\text{C}_2\text{H}_2$ ) as the carbon carrier, conducted in a multi-stage pulse mode with precisely controlled partial pressure [2]. As illustrated in Fig. 4, the experiments were performed using a DB-644 vacuum low-pressure carburizing furnace. A total of 18 carburizing schedules were designed by combining variations in carburizing temperature (1050 °C and 980 °C, 3 h), secondary austenitizing temperature (1000 °C, 1020 °C, and 1040 °C for 40 min), and tempering conditions (160 °C, 200 °C, and 240 °C in a box-type resistance furnace,

**Table 1**  
Chemical compositions of the three test steels (in wt.%).

| Sample   | C    | Si   | Mn   | P    | Cr    | Fe   |
|----------|------|------|------|------|-------|------|
| Sample A | 0.07 | 1.02 | 1.06 | \    | 11.52 | Bal. |
| Sample B | 0.18 | 0.91 | 0.99 | \    | 13.31 | Bal. |
| Sample C | 0.24 | 0.98 | 1.05 | \    | 11.92 | Bal. |
| Sample D | 0.35 | 0.62 | 0.8  | 1.04 | 12.84 | Bal. |



**Fig. 3.** Illustration of VMD Algorithm Parameters. (b) and (c) represent the VMD decomposition of the wear curve (a), covering high-frequency (b)(d) and low-frequency (c)(e) modes and their corresponding intensities. (d) and (e) show the frequency signals corresponding to the two modes.



**Fig. 4.** Vacuum Carburization Heat Treatment Experimental Data Collection: Vacuum carburization heat treatment process route and corresponding parameters.

followed by air cooling). These schedules were applied across multiple steel grades. Together, they yielded a dataset of 2,744 instances, each consisting of a complete set of process parameters, microstructural images, and experimentally measured mechanical properties.

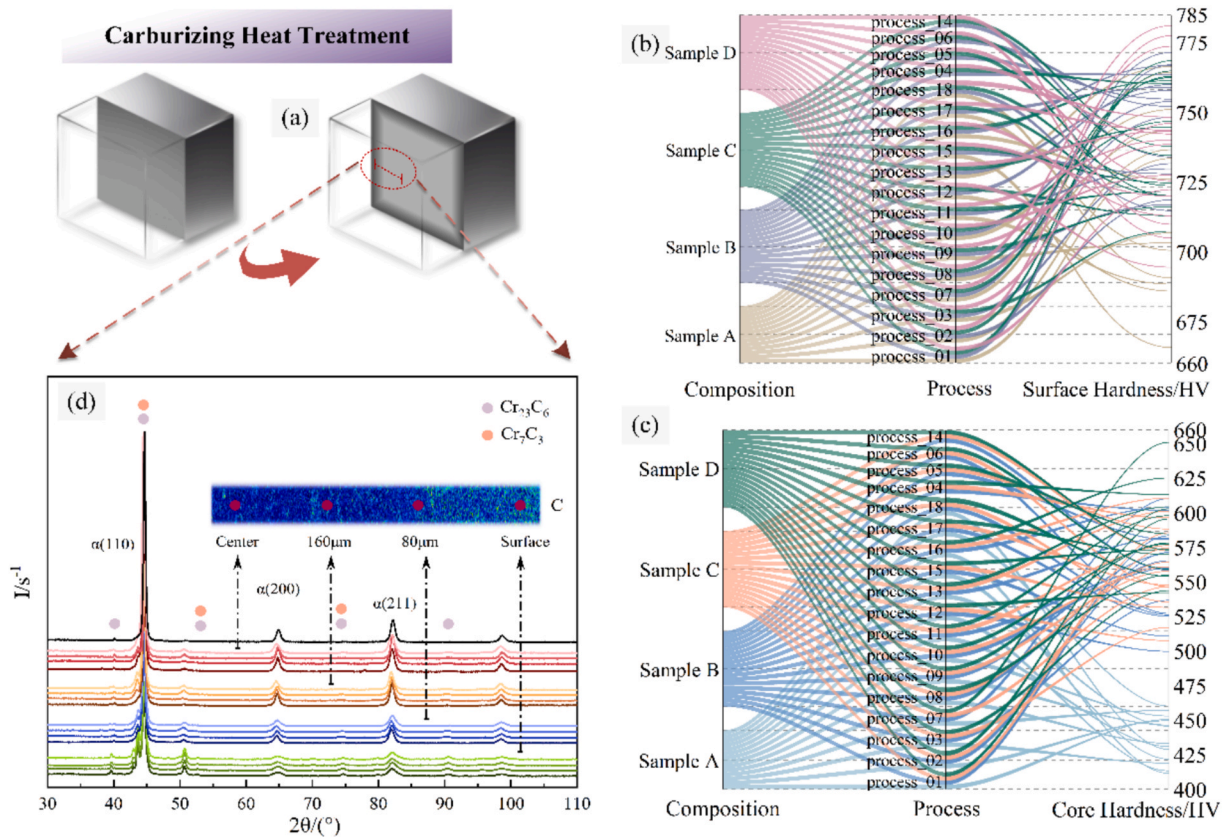
### 3.2. Multimodal Dataset Creation

#### 3.2.1. Metallographic Imaging and Property Characterization

To construct a more systematic relationship of “process parameters – composition – microstructure – mechanical properties” for carburized steel, this study established a multimodal dataset that includes different compositions, process parameters, microstructural images, and corresponding mechanical properties. Among them, the acquisition of

microstructural information was a key component of the dataset.

Each sample was embedded in resin for polishing. The grinding process began with 240-grit paper to flatten the surface, followed by 400, 600, and 800-grit to remove finer defects. Final pre-polishing steps were performed with 1000, 1500, and 2000-grit papers to ensure surface smoothness. Subsequently, samples were polished with 3  $\mu\text{m}$  diamond suspension to eliminate scratches. For microstructural observation, the polished surface was etched using a solution of 4 % nitric acid, 4 % hydrofluoric acid, and 92 % deionized water for several seconds. In this study, a BH200M metallographic microscope was used to capture metallographic images of the carburized surface and carburized gradient surface (with an image size of  $2592 \times 1944$  pixels). The extracted metallographic image features were utilized as one modality in the



**Fig. 5.** Multimodal Dataset Analysis. (a) Cross-section of heat-treated samples and schematic of the carburized layer. (b) Surface hardness distribution of samples (660 ~ 785 HV). (c) Core hardness distribution of samples (400 ~ 650 HV). (d) XRD spectra and EDS data of the carburized layer, covering detection results from the surface to the core for four types of materials.

multimodal prediction of wear performance and hardness distribution. The metallographic images reflect the structure of various phases, phase morphology, and other latent features of the samples. Due to the diffusion of carbon atoms within the matrix, the micrographs of the carburized gradient surface display a transition region from the edge to the core, as shown in Fig. 5 (a). In the carburized layer, martensite is prone to etching and appears dark, while residual austenite is resistant to etching and appears white, with carbides present in the white areas. There is a carbide enrichment at the sample edges, significantly enhancing the hardness of the near-surface region.

Hardness curves were obtained using an HVW-1000 ZT Vickers hardness tester. The sample size was  $10 \times 10 \times 10$  mm, and after mounting, indentation measurements were conducted on the carburized gradient surface from the surface towards the core at 0.1 mm intervals until the hardness gradient leveled off. Friction curves were obtained using a BMFWG-02 friction and wear testing machine. The sample size was  $10 \times 10 \times 20$  mm, with a counter ball of 6 mm diameter Si<sub>3</sub>N<sub>4</sub>, and a load of 150 N was applied for 10 min of dry friction under room temperature conditions.

### 3.2.2. Data Augmentation and Dataset Construction

The dataset was constructed and data augmentation was performed prior to training. The metallographic images of carburized samples collected using the aforementioned methods were categorized into two types: (1) images of the carburized surface, serving as one modality for predicting surface wear performance, and (2) images of the carburized gradient surface, used as a modality for predicting hardness distribution. To enhance the dataset, two augmentation techniques were employed: rotation and cropping. Images were rotated at 90° intervals to increase orientation diversity, while cropping was used to standardize the image size to  $370 \times 243$  pixels. Importantly, because accurate prediction of the hardness gradient requires preserving the full spatial features of the carburized gradient surface, only rotation was applied to these images, and cropping was omitted. For model input, the cropped images were directly resized to  $224 \times 224$  pixels and fed into the microstructure encoder. In contrast, the full-resolution gradient surface images ( $2592 \times 1944$  pixels) were divided into multiple non-overlapping  $224 \times 224$  patches using a sliding window approach, ensuring complete spatial coverage.

The augmented dataset (hereafter referred to as the dataset) was divided into training, validation, and test sets in a 7:2:1 ratio, comprising 1922, 548, and 274 samples, respectively. The training set was used to optimize the model's weights and biases, while the validation set assisted in tuning hyperparameters during training. The test set was excluded from the training process and served exclusively to evaluate the model's predictive accuracy and other performance metrics.

### 3.2.3. Dataset Analysis

After vacuum carburization treatment with different combinations of process temperatures, a carbon concentration gradient distribution approximately  $2.2 (\pm 0.6)$  mm deep (from the surface) was formed on the sample surface, with an effective carburizing depth of  $0.9 (\pm 0.15)$  mm. This covers a hardness range of 400 ~ 780 HV from high-carbon to low-carbon regions, adequately encompassing the typical C concentration and hardness distribution observed in vacuum carburization heat treatment, and demonstrating acceptable data diversity. Influenced by different carbon concentrations in the matrix, the core hardness of vacuum-carburized samples exhibits a distinct distribution pattern. Samples with lower initial C content have lower core hardness after vacuum carburization, while samples with higher initial C content have higher core hardness after treatment. This pattern is also slightly reflected in the surface hardness of the samples, though it is less pronounced, as shown in Fig. 5 (b-c). Additionally, representative samples were randomly selected from the sample library for X-ray diffraction (XRD) analysis, ensuring that the full hardness range was covered (as illustrated in Fig. 5 (d)). The results indicate that with the increase of

carbides, the material's hardness improves and wear performance is enhanced, aligning well with existing theoretical research.

### 3.3. Evaluation Metrics and Error Estimation

To evaluate model performance across different prediction tasks, we adopted both conventional and task-specific metrics. For hardness curve prediction, each sample yields a predicted curve along the depth direction. Mean Absolute Error (MAE) and  $R^2$  were calculated point-wise along each curve, and then averaged across all test samples to obtain overall performance scores. This approach allows fine-grained evaluation while maintaining consistency across sample-level predictions.

In addition, to more intuitively demonstrate the effect of multimodal input on prediction quality, we defined a simple Accuracy metric tailored for hardness prediction. Specifically, for each sample, if the absolute difference between every predicted point and the corresponding ground truth value on the hardness curve is less than 10 HV, the sample is considered a correct prediction. This classification-style metric complements MAE and  $R^2$  by offering an interpretable threshold-based measure of prediction reliability.

## 4. Results and discussion

### 4.1. Correlation Between Multimodal Parameters and Mechanical Properties

During the vacuum carburization process, the final mechanical properties of the material are governed by a complex interplay among microstructure, composition, and process parameters. Metallographic images capture internal grain boundaries, precipitated phases, and phase distributions; compositional data reflect the chemical makeup; and process parameters describe the carburizing conditions, including temperature, pressure, and time.

Changes in hardness during carburization are closely associated with phase distribution, carbide morphology, and other microstructural features. These characteristics, visible in metallographic images, form the basis for predicting the hardness distribution across the carburized layer. Fine, uniformly distributed grain boundaries promote layer homogeneity, enhancing both hardness and wear resistance. In contrast, excessive or unevenly distributed precipitates can cause localized hardness fluctuations, negatively impacting mechanical performance. Multimodal data from cross-sectional metallographic images are used to predict hardness curves (Fig. 6 (a)(c)), while data from surface images are employed to predict wear curves (Fig. 6 (b)(d)).

Material composition is another key factor. Carbon concentration significantly affects both hardness and wear resistance: higher concentrations aid in layer formation and hardening but can lead to embrittlement and reduced toughness if excessive. Alloying elements such as chromium, nickel, and molybdenum also influence the carburization process. For example, chromium improves martensite stability, corrosion resistance, and hardness. When properly balanced, alloy additions enhance surface hardness while maintaining overall mechanical integrity.

Process parameters—particularly temperature, time, and pressure—directly influence carburized layer formation. Higher temperatures and longer durations accelerate carbon diffusion, increasing layer depth and uniformity. However, excessive conditions may cause over-carburization and degrade toughness. Additionally, maintaining appropriate vacuum levels and controlling the carburizing atmosphere is essential for minimizing impurities and ensuring high-quality, uniform layers.

### 4.2. Multi-input prediction for Hardness

To further assess the robustness and generalization capability of our multimodal prediction model, we tested it on 300 samples made from

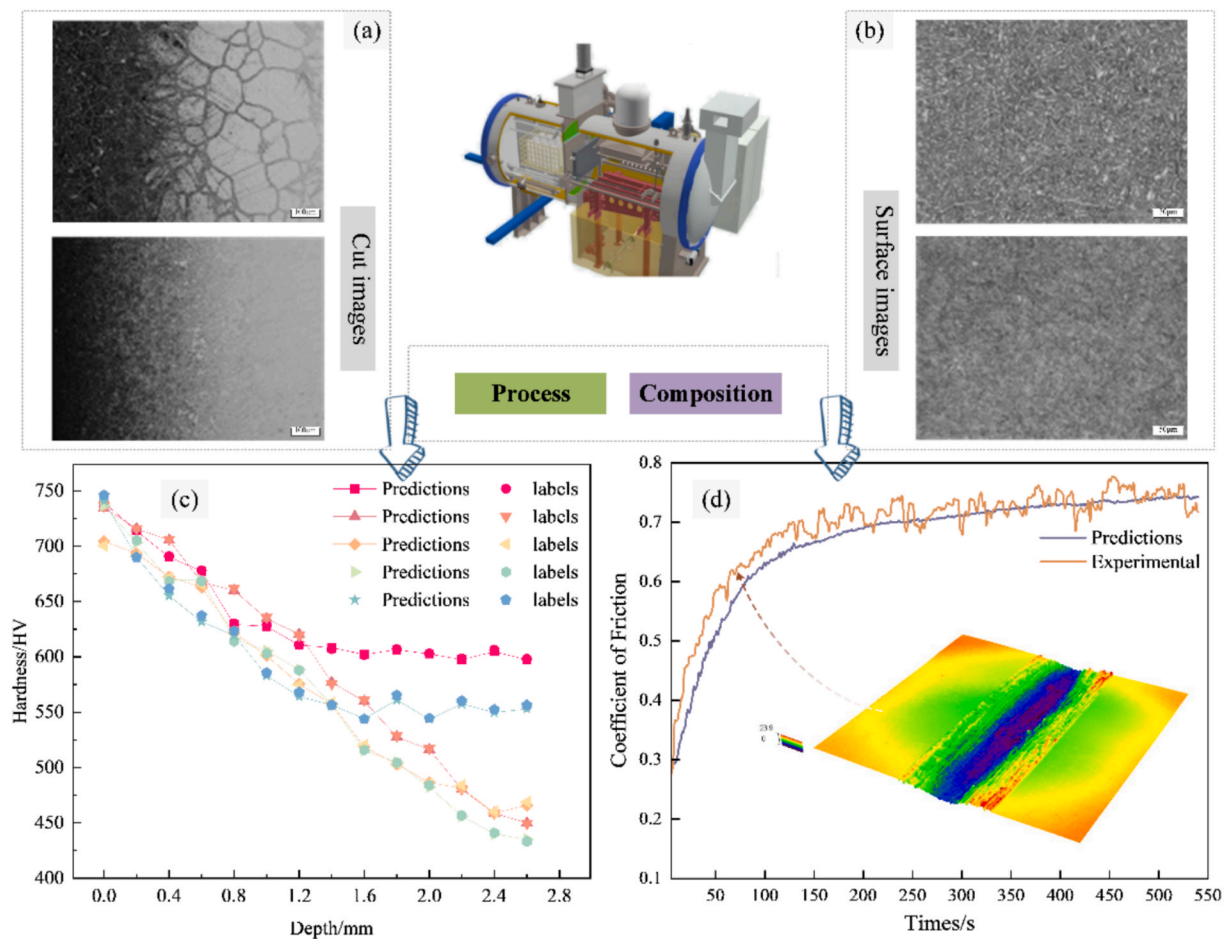


Fig. 6. Correlation Between Multimodal Parameters and Mechanical Properties. (a) Two different metallographic structures used to predict surface wear coefficient and carburized layer hardness distribution curves. (b) Predictions of Coefficient of Friction. (c) Predictions of Hardness. (d) Model Predictions for the Testing Sets.

steels different from those used in training, specifically 20CrMnTi, 20CrMo, and 42CrMo. Compared with the training alloys, these validation steels exhibit a notably lower chromium content. Nevertheless, there are also important commonalities: both training and validation alloys are Fe-based steels with overlapping carbon and manganese ranges, which are well known to govern carburization kinetics, hardness, and microstructural evolution. These shared metallurgical features provide a sound basis for the model to generalize beyond the exact chemistry of the training set. Importantly, the validation data were not involved in training, enabling a comprehensive assessment of the model's predictive reliability.

Fig. 7 shows the hardness curve predictions in the 400–780 HV range, indicating that the proposed model can accurately predict after training. Whether using only metallographic images or all modal data including microstructural images, material composition, and process parameters, the model achieved accurate prediction results. In comparison, the model with multimodal input achieved a MAE of 5.23 HV and an  $R^2$  of 0.98, significantly outperforming single-modal input. This demonstrates that multimodal data can more closely approach the ideal prediction line with a slope of 1, and the small fluctuations in prediction bias indicate a high correlation between the predicted results and the actual values.

We used models with single-modal, two-modal, and three-modal inputs and presented the accuracy and MAE for seven combinations (as shown in Fig. 8). Among them, the test groups using only material composition or process parameters showed large MAE due to insufficient information and the results were not of reference value. In contrast,

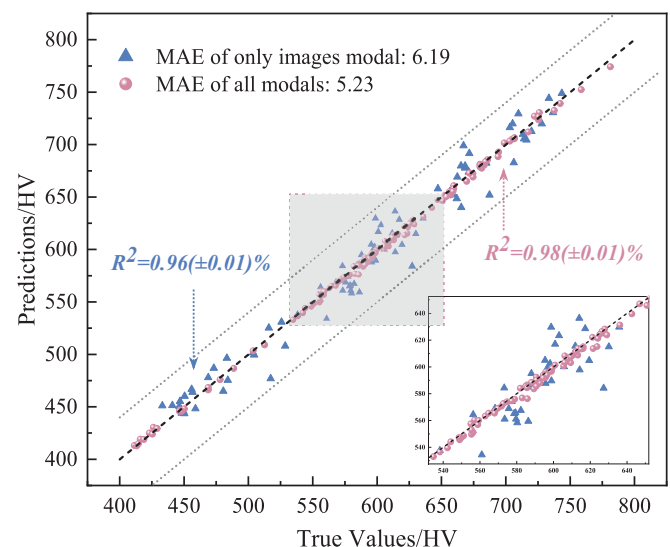


Fig. 7. Prediction results of the multimodal model on the hardness distribution in the test set.

models incorporating microstructural image data improved accuracy to  $91(\pm 0.014)\%$ . Models using only microstructural images and process parameters, or microstructural images and material composition achieved accuracies of  $83(\pm 0.012)\%$  and  $85(\pm 0.037)\%$ , respectively.

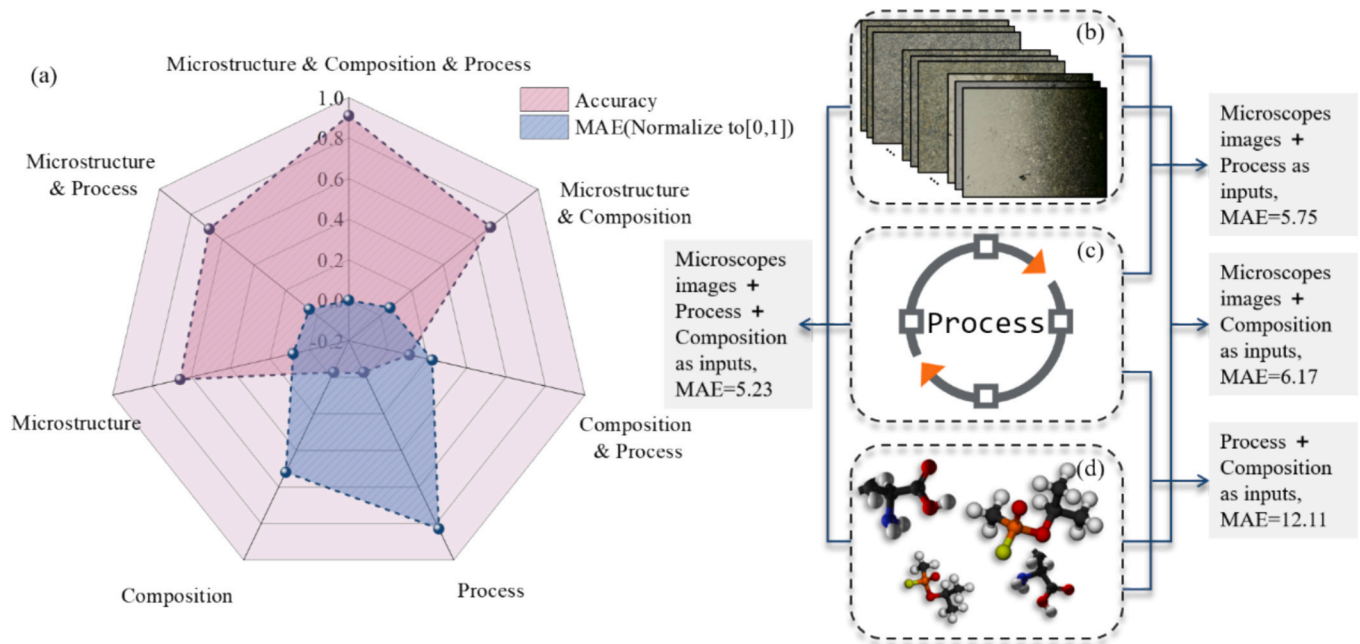


Fig. 8. Comparison of prediction results from different data modal.

Compared to single-modal input, the inclusion of any additional modality significantly improved prediction accuracy, further demonstrating the importance of material composition and process parameters in predictions.

#### 4.3. Multi-input prediction for Friction

Fig. 9 shows the deviation between the predicted wear curve results and the actual values. Unlike the hardness predictions shown as smooth curves in Fig. 6(d), the friction prediction results in Fig. 9 are displayed as individual time-wear data points, which makes local deviations and transient noise more visually prominent. It can be seen that, the model's MAE and R2 for wear curve prediction are 0.04 and 0.74. This indicates that the wear curve prediction task is more complex, with a larger number of prediction points, and the wear curve is more susceptible to

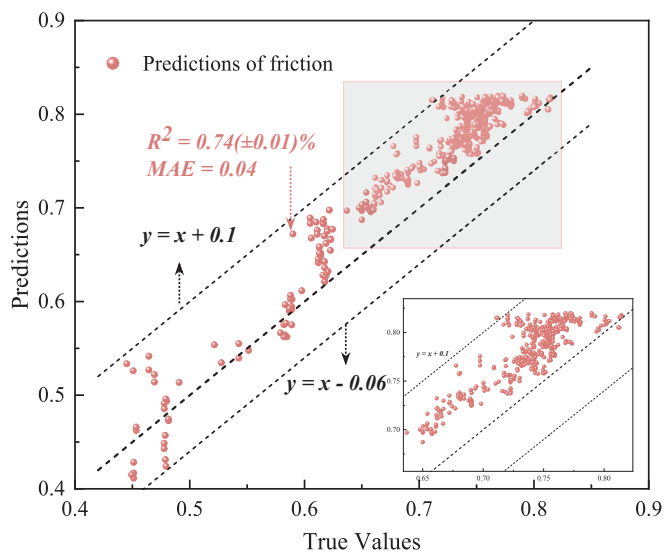


Fig. 9. The prediction results of the multimodal model for friction performance on the test set.

environmental and equipment noise during data collection, which affects the accuracy of the prediction results. Compared to hardness curves, wear curves may contain more variability and uncertainty, making it difficult for the model to accurately capture underlying patterns during the learning process.

As shown in Fig. 10, the raw wear data contains multiple frequency components and random noise, which can obscure meaningful patterns and reduce prediction accuracy. In contrast, the VMD-processed data separates the wear signal into low- and high-frequency components, effectively filtering out environmental and equipment noise. This decomposition enables the model to focus on physically meaningful trends in the friction response. In particular, we analyzed the model's performance on each individual intrinsic mode function (IMF). While IMF2 naturally has a small MAE due to its low amplitude, the more important low-frequency component, IMF1—reflecting the main wear progression—also exhibits a significant reduction in MAE after VMD preprocessing. This suggests that VMD not only improves data quality but also enhances the model's ability to capture the underlying physical behavior, leading to more robust and interpretable predictions.

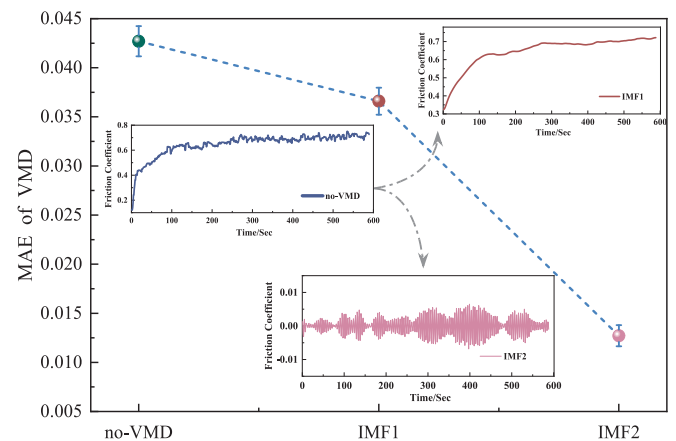


Fig. 10. Comparison of Model Prediction Performance Using Raw and VMD-Processed Wear Data.

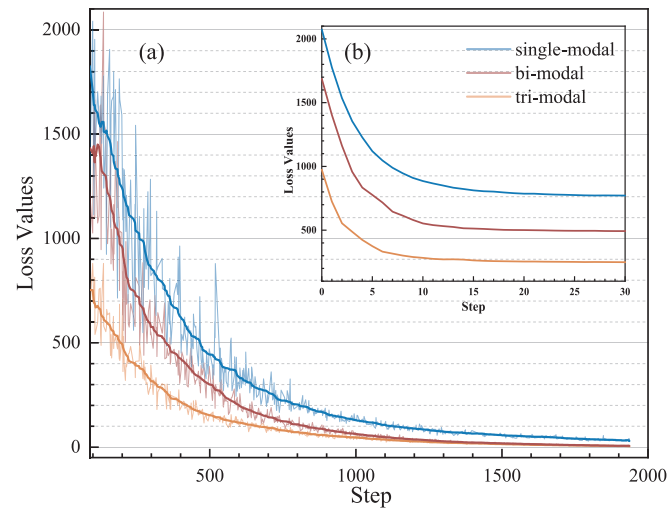


Fig. 11. Convergence process of model training. (a) model training process, (b) validation process.

4.4. Convergence and Stability Analysis

Fig. 11 shows the convergence behavior of the proposed Transformer-based multimodal learning framework under different modality settings. Specifically, we conducted ablation experiments using three configurations: single-modal (microstructure only), bi-modal (microstructure + parameters), and tri-modal (microstructure + parameters + process conditions).

As illustrated in Fig. 11 (a), which depicts the training loss curves, the tri-modal model demonstrates the fastest convergence and achieves the lowest final training loss, followed by the bi-modal and single-modal versions. Moreover, the tri-modal loss trajectory is smoother and exhibits less oscillation, indicating stronger robustness to batch-level variations and improved training stability. Fig. 11(b) presents the corresponding validation loss curves during the early training phase. It highlights the model’s generalization behavior across the first 30 steps. The tri-modal configuration consistently outperforms the others in validation loss, suggesting that incorporating complementary modalities facilitates faster and more stable learning not only on the training set but also on unseen data. These results collectively indicate that multimodal fusion enhances the optimization process by providing richer feature representations. The Transformer architecture is able to exploit cross-modal interactions effectively, leading to improved convergence speed, better generalization, and overall model stability.

4.5. Comparative Evaluation of Multimodal Learning Strategies

In this study, we proposed a multimodal learning framework combined with MLP regression to predict the mechanical properties of

carburized steel after vacuum carburization. To further assess the effectiveness of this approach, we compared it against several commonly used multimodal learning strategies, including recent vision-language and multimodal Transformer-based models such as CLIP, ViLT, UniT, and MMBT, as well as traditional CNN-based baselines (e.g., ResNet and VGG).

As summarized in Table 2, our method significantly outperforms all comparative models in terms of both MAE and  $R^2$ , demonstrating superior accuracy and generalization performance. Specifically, we implemented and adapted several representative multimodal learning strategies for performance comparison, based on open-source implementations with minor modifications to fit our regression task: CLIP was adapted by replacing its original natural language inputs with structured material composition and process parameters, reformatted as pseudo-text (e.g., “C: 0.25, Cr: 1.2...”). We retained the original ViT image encoder and encoded both modalities separately. The resulting embeddings were projected into a shared space, concatenated, and fed into an MLP for regression prediction; ViLT was modified to directly concatenate embedded microstructural image patches and numerically tokenized parameter embeddings as a single input sequence. This sequence was passed into a unified Transformer encoder, whose output was connected to an MLP for final prediction; UniT incorporated both image and parameter modalities by projecting them into a shared latent space. The fused sequence was then fed into a multimodal Transformer encoder, and the output corresponding to the regression task token was passed through an MLP for predicting hardness and wear properties; MMBT adopted an early fusion strategy, where microstructural images were encoded using a ResNet-50 backbone, and structured parameter inputs were embedded and concatenated with the image features. The combined sequence was then processed through a BERT-style encoder, and the output token representation was connected to an MLP for downstream regression.

Among these, MMBT achieved relatively strong performance (MAE = 6.75,  $R^2$  = 0.928), validating the importance of structured early fusion. However, our full-fusion Transformer model further improved prediction accuracy, achieving the lowest MAE of 5.23 and the highest  $R^2$  of 0.980, due to its capacity to jointly model long-range interactions across microstructural and parametric features.

Furthermore, Fig. 12 illustrates the metallographic feature maps extracted from various convolutional depths of ResNet-50. While shallow layers are able to capture basic visual cues, deeper layers tend to produce abstract representations that fail to retain fine-grained structural information of the material. In contrast, our multimodal Transformer-based model effectively preserves these details through global self-attention, which is particularly important when dealing with the spatially heterogeneous patterns of material microstructures. Overall, the results of this comparative study indicate that Transformer-based multimodal learning strategies, particularly those leveraging full fusion across modalities, are more effective than conventional CNNs or early fusion architectures. The proposed framework provides improved flexibility, accuracy, and robustness for predicting mechanical properties in complex materials science tasks.

Table 2  
Comparison of the Performance of Different Multimodal Learning Strategies.

| Multimodal Learning Strategies | Modalities Used              | Type               | MAE   | $R^2$ |
|--------------------------------|------------------------------|--------------------|-------|-------|
| CLIP [40]                      | Image + Parameters + Process | Vision-Language    | 8.45  | 0.890 |
| ViLT [41]                      | Image + Parameters + Process | Multimodal Trans   | 7.82  | 0.905 |
| UniT [42]                      | Image + Parameters + Process | Multimodal Trans   | 7.10  | 0.920 |
| MMBT [43]                      | Image + Parameters + Process | Early Fusion       | 6.75  | 0.928 |
| ResNet-50 [44]                 | Image only                   | CNN baseline       | 10.49 | 0.844 |
| ResNet-18                      | Image only                   | CNN baseline       | 13.90 | 0.803 |
| VGG-16 [45]                    | Image only                   | CNN baseline       | 12.61 | 0.750 |
| ResNet-50 + MLP                | Image + Parameters + Process | Early Fusion       | 7.85  | 0.882 |
| Our Method                     | Image + Parameters + Process | Ours (Full Fusion) | 5.23  | 0.980 |

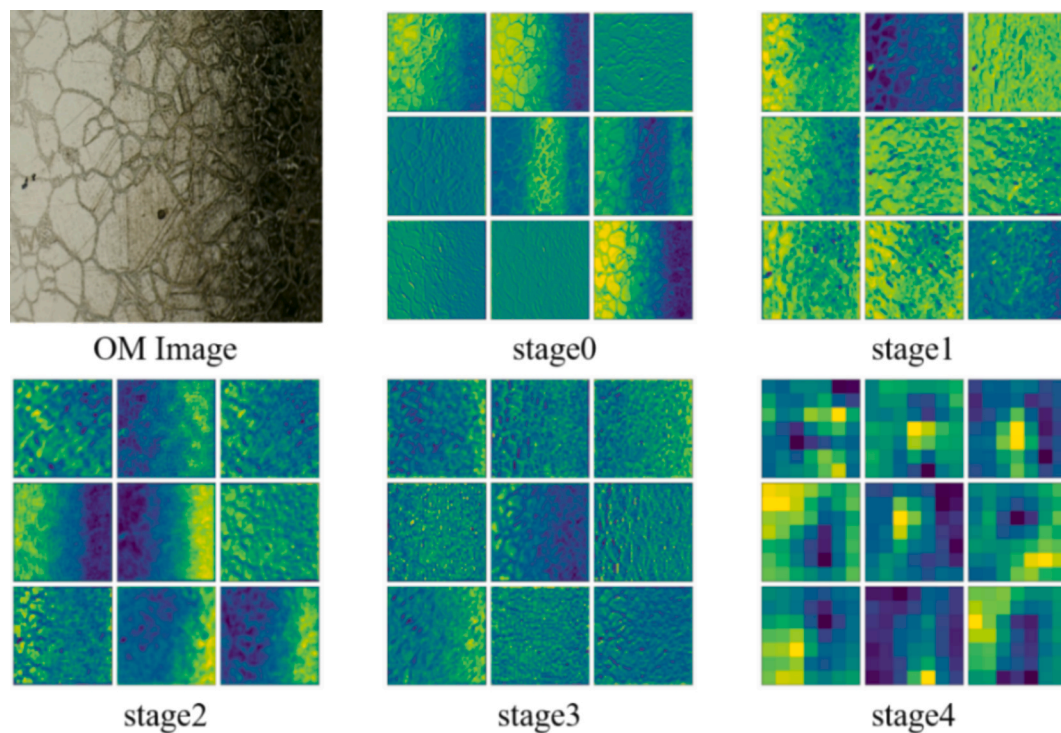


Fig. 12. Metallographic Feature Maps Extracted from Different Depth Convolutional Layers.

#### 4.6. Discussion of Error Sources and Limitations

The sources of error in this study primarily include sample preparation artifact interference and instrument system and environmental disturbance errors. Regarding sample preparation artifact interference, scratches generated during the polishing process may affect the model's judgment of carbide distribution, leading to deviations in local hardness predictions. Concerning instrument system and environmental disturbance errors, factors such as temperature fluctuations and high-frequency noise during friction testing may impact the model's predictive accuracy and reliability.

Additionally, the lack of multi-scale coupled data presents a significant challenge for the current model, especially regarding the transition between micro and meso scales. Existing studies often use metallographic images as inputs but do not integrate nanoscale information such as Electron Backscatter Diffraction (EBSD) grain boundary features or Transmission Electron Microscopy (TEM) dislocation structures. This limitation arises partly from the high costs associated with EBSD or TEM analysis, which offer relatively limited improvements in model accuracy. On the other hand, these high-cost detection methods contradict the initial simplicity and practicality envisioned for the model.

Furthermore, our proposed multimodal learning model is not intended to replace physics-based models, such as those grounded in Fick's second law or implemented in FEM software (e.g., Deform, Dante, Cosmap), which provide highly interpretable and accurate simulation results including carbon diffusion evolution and phase transformation kinetics. Rather, our method complements such approaches by offering data-driven predictions based on complex empirical inputs (e.g., microstructure images) that are difficult to parameterize in physics-based formulations. While our model does not simulate spatiotemporal evolution explicitly, it captures the final structure-property mapping, especially useful in high-throughput, diverse experimental settings. Future work may consider hybrid frameworks integrating

synthetic FEM-generated data or incorporating physics-informed neural networks (PINNs) to enhance interpretability and extrapolation capability.

#### 5. Conclusions

In this study, we proposed a novel multimodal learning approach for predicting the mechanical properties of carburized steel after vacuum carburizing. By integrating microstructure images, material composition data, and process parameters, our method effectively utilizes the complementary information from multiple data modalities to enhance the accuracy of hardness and wear predictions. Our experimental results demonstrated that the multimodal model outperforms single-modal approaches, achieving significant improvements in prediction accuracy. Specifically, the inclusion of microstructure images, along with material composition and process parameters, led to a substantial reduction in mean absolute error (MAE) and an increase in R2 values, highlighting the importance of leveraging multiple data sources for enhanced predictive performance. Additionally, the application of Variational Mode Decomposition (VMD) for processing wear curves effectively mitigated noise interference, further improving the model's prediction of wear performance.

Despite these promising results, there are still challenges that need to be addressed. Future work could explore the incorporation of more advanced microstructural information, such as electron backscatter diffraction (EBSD) or transmission electron microscopy (TEM) data, to further refine the model's predictions. Additionally, the development of more efficient algorithms to handle the computational complexity of multimodal data fusion could make this approach more applicable to industrial settings.

Overall, this study provides a robust framework for predicting the mechanical properties of materials through the integration of multiple data sources, demonstrating the potential of multimodal learning in

materials science and paving the way for future advancements in predictive modeling for material design and processing optimization.

### CRedit authorship contribution statement

**Xuefei Wang:** Writing – review & editing, Writing – original draft, Software. **Shijie Zhang:** Software. **Di Jiang:** Writing – original draft. **Wei Yu:** Writing – review & editing. **Yihao Zheng:** Software. **Chunyang Luo:** Writing – review & editing, Visualization. **Haojie Wang:** Methodology. **Zhaodong Wang:** Funding acquisition.

### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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### Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.matdes.2025.114800>.

### Data availability

The datasets and code that support the findings of this study are openly available in the GitHub repository Vacuum-Carburized-prediction at: <https://github.com/NotUrNeighborMrWang/Vacuum-Carburized-prediction.git>

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